

Day -10 : Review and practice practical

1. Executive Summary & Overview :

This document provides a complete, beginner-friendly guide to building and configuring a basic Wireless

Home Network using Cisco Packet Tracer. The goal of this practical lab is to establish seamless local

connectivity between three Personal Computers (PCs) and a Wireless Home Router using static IP

addressing.

Network Addressing Scheme

All devices are assigned static IPv4 addresses within the 192.168.10.0/24 subnet mask range:

Device Name	Interface	IP Address	Subnet Mask	Default Gateway
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Wireless Router	LAN / Wireless	192.168.10.1	255.255.255.0	N/A
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(Gateway Device)

PC1	FastEthernet0 / Wireless	192.168.10.2	255.255.255.0	192.168.10.1
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PC2	FastEthernet0 / Wireless	192.168.10.3	255.255.255.0	192.168.10.1
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PC3	FastEthernet0 / Wireless	192.168.10.4	255.255.255.0	192.168.10.1
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2. Network Topology Structure :

PACKET TRACER TOPOLOGY DIAGRAM (HOMEWIFINETWORK.PKT)

Router

Wireless Router

192.168.10.1

PC1

192.168.10.2

PC2

192.168.10.3

PC3

192.168.10.4

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3. Step-by-Step Configuration Guide :

1 Create Project & Add Network Devices

Follow these instructions to set up the workspace in Cisco Packet Tracer:

Launch Cisco Packet Tracer.

Go to the top menu and click File > Save As.... Save the project as HomeWiFiNetwork.pkt.

From the bottom-left Device Selection Box, navigate to Wireless Devices and drag a Wireless Router (e.g., WRT300N or standard Home Router) onto the logical workspace.

Navigate to End Devices and drag 3 PCs (PC1, PC2, and PC3) onto the workspace.

(Optional Wireless Module setup): If connecting via Wi-Fi, click each PC, power it off in the Physical

tab, replace the Ethernet card with a WMP300N wireless module, and turn the PC back on.

Alternatively, connect using Copper Straight-Through cables from the PC's FastEthernet port to

the Router's Ethernet LAN ports.

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2 Configure Router & Static IP Addresses on PCs

Set up static IP parameters on each end device to ensure they reside on the same network subnet:

A. Router IP Setup

Click on the Wireless Router > GUI Tab > Basic Setup. Set the LAN IP Address to 192.168.10.1 and

Subnet Mask to 255.255.255.0. Click Save Settings.

B. PC IP Configuration Windows

PC1 Configuration Window (Desktop > IP Configuration)

IP Configuration: ☐ DHCP ☒ Static

IPv4 Address: 192.168.10.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

PC2 Configuration Window (Desktop > IP Configuration)

IP Configuration: ☐ DHCP ☒ Static

IPv4 Address: 192.168.10.3

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

PC3 Configuration Window (Desktop > IP Configuration)

IP Configuration: ☐ DHCP ☒ Static

IPv4 Address: 192.168.10.4

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.10.1

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3 Test Network Connectivity (Ping Verification)

To verify that all devices communicate successfully, open the Command Prompt on PC1 (click PC1 >

Desktop tab > Command Prompt) and perform ping tests to the other devices.

1. Ping Test from PC1 to Wireless Router (192.168.10.1)

Packet Tracer PC Command Line v1.0

```
PC1> ping 192.168.10.1
```

Pinging 192.168.10.1 with 32 bytes of data:

```
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
```

```
Reply from 192.168.10.1: bytes=32 time=0ms TTL=255
```

```
Reply from 192.168.10.1: bytes=32 time=0ms TTL=255
```

```
Reply from 192.168.10.1: bytes=32 time=0ms TTL=255
```

Ping statistics for 192.168.10.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

2. Ping Test from PC1 to PC2 (192.168.10.3)

```
PC1> ping 192.168.10.3
```

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=0ms TTL=128

Reply from 192.168.10.3: bytes=32 time=0ms TTL=128

Reply from 192.168.10.3: bytes=32 time=0ms TTL=128

Reply from 192.168.10.3: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.10.3:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

3. Ping Test from PC1 to PC3 (192.168.10.4)

PC1> ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time=0ms TTL=128

Reply from 192.168.10.4: bytes=32 time=0ms TTL=128

Reply from 192.168.10.4: bytes=32 time=0ms TTL=128

Reply from 192.168.10.4: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.10.4:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Verification Summary

With 0% packet loss across all tests, network layer connectivity is successfully established across all three

PCs and the Wireless Router within the 192.168.10.0/24 network.

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4. Quick Troubleshooting Tips :

Request Timed Out: Double-check that all PCs are assigned IP addresses in the same range

(192.168.10.x) and have the exact same Subnet Mask (255.255.255.0).

Wireless Connection Not Establishing: Ensure the SSID on the wireless router matches the SSID in the

PC's wireless adapter settings (Default is usually 'Default' or 'HomeNetwork').

Wrong Gateway: Ensure each PC's Default Gateway is set precisely to the Wireless Router's IP address

(192.168.10.1).